WORAPON SANGSASRI

Fourth-year Electrical Engineering Student · AI HealthTech and Embedded Systems
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PROFILE

AI HealthTech and embedded-systems engineer focused on assistive motion sensing, smart-home control, and safe neuro-symbolic AI agents for elderly and wheelchair users. My flagship platform, **WheelSense**, brings together indoor positioning, motion tracking, smart-home control, and an AI assistive-care layer; **EaseAI** is the nursing-home intelligent-assistance track within the same WheelSense platform.

FEATURED RECOGNITION

Thailand National Winner – AI Ready ASEAN Youth Challenge 2026 May 2026
WheelSense, Singapore Regional Showcase – Represented Thailand and demonstrated the project to the **President of Singapore**.

Honorable Award – 2026 Best AI Awards, Taiwan Apr 2026
EaseAI within WheelSense – Recognized in the AI Student Division for combining AI motion tracking, neuro-symbolic assistance, and nursing-home smart-environment support.

EDUCATION

Thammasat University 2022 – Present
School of Engineering, B.Eng. in Electrical Engineering GPA: 3.72

Kasetsart University Laboratory School 2009 – 2021
Center for Educational Research and Development, Science-Mathematics Program GPA: 3.87

WORK EXPERIENCE

Automation Engineer Intern Jun 2025 – Jul 2025
Charoen Pokphand Group Co., Ltd. (CP CIXI Park), Zhejiang, China

- Selected for a prestigious international internship program to spearhead the integration of advanced automation systems within a large-scale industrial environment.
- Acquired comprehensive expertise in optimizing production processes and engineering complex solutions.
- Demonstrated exceptional adaptability and cross-cultural communication skills, fostering a global engineering perspective.
- Honored with the **Special Contribution Award** for delivering outstanding results and technical excellence.

Student Association Member Aug 2024 – Jun 2025
Faculty of Engineering, Thammasat University

- **Co-founder, TSE OPEN HOUSE 2024:** Orchestrated strategic planning and logistics to successfully showcase engineering programs.
- **Line Bot Developer:** Engineered a comprehensive communication system for news dissemination, student welfare, and consulting, significantly enhancing operational efficiency.
- **Audiovisual Staff:** Directed technical operations for sound, video, and live streaming, ensuring seamless event execution.
- Managed budget allocation and coordinated agendas to drive successful student activities.

PROJECTS

WheelSense: AI Motion Tracking System with Smart Home Controller for Elderly and People with Disabilities Jul 2025 – Present
Faculty of Engineering, Thammasat University

- Developed an assistive HealthTech platform combining **RSSI fingerprint localization**, motion tracking, smart-home control, and caregiver monitoring for elderly and wheelchair users.
- Built a **neuro-symbolic AI pipeline** using **Gemma 4 E2B**, rule-based safety logic, **Model Context Protocol (MCP)**, and Home Assistant for auditable smart-home actions.
- Engineered a scalable **Next.js 16**, **FastAPI**, and **Docker** architecture for real-time dashboards, AI command orchestration, and care workflow support.

RESEARCH PUBLICATION

Accepted ECTI-CON Paper: BLE-Based Indoor Motion Tracking with ML and LLMs May 2026
ECTI-CON 2026, Track 1: AI, Data & Intelligent Systems · Paper ID: 1571268966

- **First author** of "Comparative Performance Evaluation of BLE-Based Indoor Motion Tracking Using Machine Learning and Large Language Models," scheduled for Pacific 1, 14:30–14:45 on 29 May 2026.
- Built an experimental BLE RSSI benchmark using **four ESP32-S3 anchors** and a **M5StickC Plus2 wearable tag** across body-orientation, boundary-transition, and multi-zone trajectory tasks.
- Compared **KNN**, **XGBoost**, **Claude Opus 4.6**, and **Gemini 3.1 Pro**; Gemini achieved **97.14%** exact-sequence accuracy and LLMs showed stronger resilience under noise and anchor-failure tests.

ADDITIONAL HONORS AND AWARDS

Winner (Grand Prize) – Thailand New Gen Inventors Award 2026 Jan 2026
National Research Council of Thailand (NRCT)

– Project: "Wheel Sense: Motion Tracking Sensor with Smart Home Controller for Wheelchair Users."
– Also received the **Popular Vote Award**.

Gold Medal – IPITEx 2026 Jan 2026
Bangkok International Intellectual Property, Invention, Innovation and Technology Exposition

– Project: "ALL Wheelchair: Motion Tracking System with Motion-Controlled Gaming."

Innovation Pitching Merit Award 2026
Hyper Interdisciplinary Conference Thailand 2026 – Awarded for the WheelSense assistive HealthTech platform.

4th Place Nationwide – TESA Top Gun Rally 2025 Nov 2025
Thai Embedded Systems Association (TESA)

– Developed an autonomous drone system for defense and surveillance using MATLAB and Raspberry Pi 5.
– Title: "The Best of the Best Embedded System Developers."

Champion (1st Place) – LabVIEW Battle Thailand 2025 Aug 2025
TECHSQUARE CO., LTD. – Solved engineering problems under time pressure using LabVIEW.

Scholarships & Academic Awards:

– Highest GPA Award, Electrical Engineering (2022, 2023).
– Mitsubishi Electric Thai Foundation Scholarship recipient.

SKILLS

Core Competencies: Internet of Things (IoT) & Smart Home, Embedded Systems Design, Industrial Automation, Machine Learning, Computer Vision, Microservices Architecture

Languages: Python, C/C++, MATLAB, LabVIEW, SQL, TypeScript, JavaScript, HTML/CSS

Frameworks & Libraries: React, Next.js, FastAPI, TailwindCSS, YOLOv8, MQTT

Tools & Platforms: Docker, Git, Node-RED, PlatformIO, nRF Connect SDK, Simulink, AutoCAD, Firebase, Home Assistant, VS Code

Soft Skills: Leadership, Cross-functional Teamwork, Complex Problem Solving, Adaptability

Languages: Thai (Native), English (TOEIC: 790)

CERTIFICATIONS

Nordic Semiconductor: Bluetooth Low Energy Fundamentals (Jul 2025), nRF Connect SDK Fundamentals (Jul 2025).

MathWorks: MATLAB Fundamentals, Simulink Fundamentals, Machine Learning Onramp.